# Open problems:
# Physical vector spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

## From the brief

1. **Completeness of the Lie algebra.** *Research.* Definition 1 admits pathological cases: a Hilbert space with the defining observables of a qubit ("too many vectors") and a Schwartz space with position as the only defining observable ("too few observables"). "Addressing this gap is an open problem"; the brief expects a "final definition" once it is solved. Partial answer (`further-results.tex`, Sec. 2): the second case is not a quantum vector space, since with position alone $\mathcal{S}(\mathbb{R})$ is not complete (`FurtherResults.not_isQuantumVectorSpace_posOnly`). The first case is one (`FurtherResults.isQuantumVectorSpace_pair`, for $\mathbb{C}^2 \otimes \mathbb{C}^2$), and irreducibility excludes it: every continuous operator that commutes with all defining observables is a multiple of the identity (`FurtherResults.not_isIrreducible_pair`). This is a form of Roberts's condition (J. Math. Phys. 7 (1966) 1097, p. 1099), and Examples 3 and 4 satisfy it (`FurtherResults.isIrreducible_schwartz`, `FurtherResults.isIrreducible_qubit`). Open: whether irreducibility is the condition the brief is looking for, and how it relates to taking the smooth vectors of an irreducible unitary group representation as $V$. (Brief, Sec. 5, "Completeness of the Lie algebra"; Sec. 6.)

2. **Lie algebras determine physical systems.** *Research.* "Two systems are physically equivalent if and only if their Lie algebras of observables are equivalent" is asserted, and is what makes QSTO:V consistent with [B]STDIST. (Brief, Theorem 5, proof; Log, "Differences", last item.)

3. **Ensembles on a quantum vector space.** *Research.* QSTO:V uses density operators $\rho$ and $S(\rho) = -\operatorname{tr}(\rho \log \rho)$, and Sec. 5 calls the set of ensembles "the most natural space to consider", but no ensemble space over $V$ is defined. In particular, whether the ensembles induce on the pure states the quotient topology of $V$, under which the pure states are second countable exactly when $V$ is, is open. (Brief, Sec. 3, QSTO:V; Sec. 5.)

4. **Field theories.** *Research.* "For a field theory, we would expect the defining observables to be at least countable"; no field-theory example of a quantum vector space is given. (Brief, Sec. 5, "Number of defining observables".)

5. **Which observables generate unitary groups.** *Research.* Since symmetric and self-adjoint coincide on $V$, "not all self-adjoint operators generate a unitary transformation", and "to

understand which ones the integrability of the system must be checked". A characterization, analogous to complete vector fields in Hamiltonian mechanics, is not given. (Brief, Sec. 5, "Symmetric vs self-adjoint".)

6. **Which changes of frame.** *Research.* The proof of Theorem 5 gets [B]WDQ from the premise that "the defining observables of A must be observables in B and vice-versa", which holds exactly when the two frames give the same topology on $V$ (`topology_eq_iff_continuous`). The brief does not say which changes of frame satisfy it. For Example 3 the change of frame $y = \tan(\frac{\pi}{2}\operatorname{erf} x)$, which the brief "Unphysicality of Hilbert spaces for quantum mechanics" uses against [B]WDQ for QSTO:H, does not: it takes a Gaussian out of $\mathcal{S}(\mathbb{R})$ (`FurtherResults.not_forall_frameUnitary_mem_range`), and the position of the second frame, carried back to the first, is not defined at that Gaussian (`FurtherResults.toLp_gaussSchwartz_not_mem_domain`; see `further-results.tex`, Sec. 1). So QSTO:V gives [B]WDQ only for a class of changes of frame that map $V$ onto itself, as Carcassi, Calderón and Aidala already require for Schwartz spaces (arXiv:2308.06669v3, Appendix, Sec. VIII). The class is not specified. The restriction cannot simply be dropped: a wave function on $\mathbb{R}$ has finite position variance in every smooth coordinate only if it vanishes outside a bounded set; then, unless it is zero, its Fourier transform does not, and the free evolution destroys the cut-off at once. This is proved in `briefs/HilbertUnphysicality/further-results.tex`, Sec. 2 (`HilbertUnphysicality.FurtherResults.not_forall_frames_freeEvolution`), and with the brief "Unphysicality of Hilbert spaces for quantum mechanics" (`HilbertUnphysicality.quantum_no_cutoff`). Open: which group of changes of frame should be part of the data of a quantum vector space, and how it relates to the Lie algebra of the defining observables. For plain changes of variable $f \mapsto f \circ g$ on $\mathcal{S}(\mathbb{R})$, those that map $\mathcal{S}(\mathbb{R})$ into itself are characterized by Galbis and Jordá (Rev. Mat. Iberoam. 34 (2018), Thm. 2.3); the unitary changes of frame carry the extra factor $\sqrt{g'}$. *Partial answer* (`further-results.tex`, Sec. 3). With a group $F$ of changes of frame acting on $V$ by linear bijections that preserve the inner product, the premise holds for every pair of frames in $F$ if and only if every element of $F$ acts continuously on $V$ (`FurtherResults.isFrameCompatible_iff_continuous`). Then every change of frame in $F$ is a unitary equivalence (`FurtherResults.IsFrameCompatible.unitaryEquiv`). It suffices that the transformed defining observables lie in the algebra of the original ones (`FurtherResults.isFrameCompatible_of_mem_algebra`); the translations of Example 3 are an instance (`FurtherResults.isFrameCompatible_translate`). The choice of $F$ remains open. (Brief, Theorem 5, proof; Log, "Problems found in the brief", last item.)

## From the formalization

None. Definition 1 and Examples 3 (in $n$ dimensions) and 4 are checked in full.